InduSense Platform — Operational User Manual

Confidence-Aware & Evidence-Grounded Predictive Maintenance System

STANDARD:	DOC CODE:	SECURITY:	VERSION:
ABB-INDUSENSE-2026	INDUSENSE-SOP-2026-V2	OT Operational Ready	2.4.0

1. System Overview & The 6-Stage Diagnostic Pipeline

InduSense is an enterprise platform engineered for industrial maintenance troubleshooting. Standard large language model RAG architectures suffer from *semantic proximity fallacies* — hallucinating irrelevant repair procedures based on keyword similarity. InduSense strictly aligns predictive sensor diagnostics with verified OEM technical standards.

> **Core Pipeline Flow:** Predict → Retrieve → Align → Validate → Adapt → Explain.

- **1. Predict:** Ingests 1 Hz or high-frequency sensor streams. Executes domain ML models (XGBoost / HistGradientBoosting / PyTorch) with isotonic/Platt calibration to calculate an uninflated condition confidence (P_c).
- **2. Retrieve:** Executes hybrid sparse (BM25) and dense neural search across indexed technical manuals and maintenance archives.
- **3. Align:** Computes the Condition-Evidence Alignment (CEA) score, matching `Component → Fault → Symptom → Action`.
- **4. Validate:** Fuses P_c, R_c, O_c (OCR score), and CEA into an authoritative reliability score. Automatically flags evidence conflicts.
- **5. Adapt:** Confidence-aware cloud router dynamically routes requests between Deterministic Light, Standard Hybrid, and Multi-Agent Heavy compute paths.
- **6. Explain & Audit:** Generates citation-grounded procedures with exact page numbers and records all sign-offs in an immutable SQLite audit trail.

2. Operational Role Matrix

InduSense enforces strict role-based authority boundaries across industrial operations:

Operational Role	Domain Responsibility	Authorized Capabilities
Lead Reliability Engineer	Plant-wide fleet reliability & governance	Full diagnostic case sign-off, engineering override authority, parameter editing, and model checkpoint retraining.
Maintenance Technician	Physical machine bay troubleshooting	Sensor telemetry ingestion, inspecting OEM defect parts, executing repair checklists, submitting field observations.
System Administrator	OT/IT infrastructure & compliance	API routing policies, inspecting immutable SQLite audit trails, managing cloud router dispatch thresholds.

3. Step-by-Step Diagnostic Procedures

3.1 Ingesting & Running Telemetry Diagnostics

1. Navigate to the **Cases** workspace view.
2. Click **Start Diagnostic Run** (`+ New Run`).
3. Select target physical asset (`CNC-M102`, `MOTOR-DRV-01`, `HYD-PRESS-P14`, or `TURBO-TF004`).
4. Choose the appropriate ML diagnostic head and upload any corresponding CSV from `demo_data_and_user_manuals/`.
5. Click **Initialize Run** to execute the full 6-stage pipeline.

3.2 Reviewing Cases & Managing Engineering Overrides

> **Safety Constraint:** If a case exhibits an *Evidence Conflict*, accepting the diagnosis requires documenting an engineering override reason of at least 15 characters. The action is recorded permanently in the tamper-evident audit trail.

The Case Workspace separates the single overall case status (`Aligned: Evidence Verified` vs `Escalated: Evidence Conflict`) from human review decisions (`Pending Sign-Off`, `Accepted [Engineering Override]`, or `Rejected`).